PYTHON ACTIVITY-BASED LEARNING

Case Study 05: Student Marks Matrix Analysis

Dr. RVR NRI Institute of Technology | Python Programming

Student Name: _____ **Roll No:** _____

Branch / Section: _____

Problem Statement

Analyse a matrix of student marks across subjects, calculate student totals and subject averages, and identify the top student.

Objectives

• Represent marks using a nested list.

• Calculate row-wise totals and column-wise averages.

• Identify the student with the highest total.

Assumptions

• Each row represents one student and each column represents one subject.

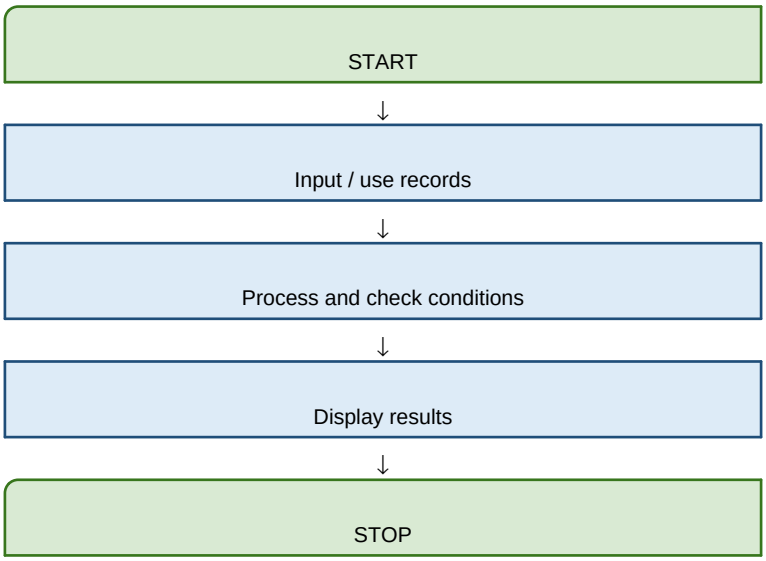
• All marks are out of 100.

• The sample contains three students and three subjects.

Algorithm

1. Create a nested list of marks and a list of student names.
2. Calculate each student's total using the row values.
3. Find the student with the maximum total.
4. For each subject column, calculate its average.
5. Display totals, subject averages and topper.

Flowchart



Pseudocode

```
START
STORE names and marks matrix
FOR each student row
    CALCULATE row total
    DISPLAY total
FIND maximum total and topper
FOR each subject column
    CALCULATE column average
    DISPLAY average
STOP
```

Python Source Code

```
names = ["Asha", "Bharat", "Charan"]
marks = [
    [80, 75, 90],
    [65, 88, 70],
    [92, 85, 95]
]

totals = []

for i in range(len(marks)):
    total = sum(marks[i])
    totals.append(total)
    print(names[i], "Total:", total)

top = totals.index(max(totals))
print("Top student:", names[top])

for col in range(3):
    total = 0
    for row in range(3):
        total += marks[row][col]
    print("Subject", col + 1, "Average:", total / 3)
```

Test Cases

Test input / condition	Expected result
Asha marks 80,75,90	Total = 245
Charan marks 92,85,95	Top student = Charan
Subject 1 values 80,65,92	Average = 79.0

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
Asha Total: 245
Bharat Total: 223
Charan Total: 272
Top student: Charan
Subject 1 Average: 79.0
Subject 2 Average: 82.67
Subject 3 Average: 85.0
```

Conclusion

The program uses nested lists and nested loops to analyse student and subject performance.